

PRIYADHARSHINI SELVARAJ

Interview Preparation Guide | SDE & Data Analyst Roles

1 Tell me about yourself.

Tests: Communication clarity, confidence, and structured thinking

I'm Priyadharshini, a final-year Computer Science student with a strong foundation in full-stack development and data analytics. Over the past year, I've built production-level systems from scratch — like a role-based Employee Attendance Management System using Spring Boot and React.js, and an AI-powered Crowd Control System that uses YOLO for real-time human detection. What sets me apart is that I didn't just complete these as assignments — I designed the architecture, solved real engineering problems, and shipped working systems. Beyond projects, I've self-learned Java, Spring Boot, React.js, Power BI, and Excel — all independently. I'm actively solving DSA problems on LeetCode and I genuinely enjoy building things that work at scale. I'm now looking for an SDE or Data Analyst role where I can apply this hands-on experience to create real impact.

◆ *Pro Tip: Structure it as: Present strength → Project proof → What you're looking for. Don't just list skills — show what you built.*

2 Why should we hire you?

Tests: Your unique value and problem-solving ability

Most candidates at my level talk about what they've learned. I can show what I've built. I've developed a full-stack attendance system with role-based access control, automated reporting, and a responsive UI — not as a tutorial clone, but as an original system solving a real-world HR problem. I've also worked on an AI system that replaces manual crowd monitoring with live YOLO detection and automated entry restriction — a real safety solution. I pick up tools fast — I self-learned Spring Boot, React.js, and Power BI without formal training. I bring a rare combination of backend engineering, frontend design, and data visualization skills. And I'm not here to just complete tasks — I'm here to understand problems deeply and build the right solutions. That's the kind of engineer I am, and that's the value I'll bring from day one.

◆ *Pro Tip: Don't be generic. Tie every claim to a specific project or achievement. Show you solve problems, not just code.*

3 What are your strengths?

Tests: Self-awareness and relevance to the role

My top strength is system thinking — I don't just write code, I think about how pieces connect. When I built the attendance system, I designed the database schema, the REST APIs, the role-based access logic, and the frontend — end to end. That ability to see the full picture and make smart design decisions early saves a lot of rework later. My second strength is fast, independent learning. I've never waited for someone to teach me — I pick a problem, find the best resources, build something real, and iterate. That's how I learned Java, Spring Boot, React.js, Power BI — all self-taught. And third — I'm genuinely curious. I don't stop at 'it works.' I ask why it works, and how to make it better.

◆ *Pro Tip: Two strong, specific strengths beat five vague ones. Always link a strength to proof.*

4 What are your weaknesses?

Tests: Honesty, self-awareness, and growth mindset

I tend to be a perfectionist when it comes to system design. Early on, I'd spend too much time refining architecture before writing a single line of code. I've learned to balance that — now I use a 'build-first, refine-later' approach. I build a working prototype quickly, then improve it based on actual feedback. That shift helped me ship projects faster without sacrificing quality. I'm also actively working on speaking up more in group settings. I've started participating in technical discussions and code reviews, and I'm getting more comfortable sharing my perspective confidently.

◆ *Pro Tip: Frame it as: Real weakness + What you did about it. Never say 'I work too hard' — it sounds scripted.*

5 Why do you want to join our company?

Tests: Research, motivation, and cultural alignment

I research companies the same way I design systems — thoroughly. What draws me to [Company Name] is [specific reason: tech stack / product impact / engineering culture]. I want to be in an environment where engineers are expected to think beyond the ticket — where you're solving meaningful problems, not just shipping features. My background in building full-stack systems and AI integrations aligns directly with the kind of work your team does. I'm not applying everywhere and hoping for a yes — I'm choosing companies where I believe I can contribute significantly and grow technically. This is one of those companies.

◆ *Pro Tip: Fill in [Company Name] and [specific reason] before every interview. Vague answers cost offers.*

6 Explain a project you worked on.

Tests: Technical depth, clarity, and ownership

I'll walk you through two projects that I'm proud of. Project 1 — Employee Attendance Management System: The problem was that most attendance systems are flat — every user sees everything, which is a privacy and access control issue. I designed a three-tier role-based access system: the Organization Head has full visibility; the Manager can view and manage employees under them; and employees can only access their own records. I built this using Spring Boot (backend), React.js (frontend), and MySQL (database), with 10+ REST API endpoints covering registration, attendance tracking, and leave management. I also automated monthly report generation, eliminating 100% of manual reporting. What makes it different from typical CRUD attendance apps: it mirrors real enterprise access control patterns, it's not just a tutorial clone, and every role serves a specific real-world need. Project 2 — AI-Based Crowd Control System: The problem was that security personnel can't manually monitor crowd density in real time in small but high-footfall areas. I used YOLO for real-time human detection, centroid tracking to monitor movement, and line-crossing logic to identify who's entering vs. exiting. The system maintains a live occupancy count, compares it against a configurable threshold, and automatically restricts entry when the limit is reached — no human decision needed in the loop. I built a real-time dashboard showing live metrics, alert thresholds, and trend analytics. What makes it different: most CCTV systems just record. This one acts — it makes autonomous safety decisions in real time, reducing reliance on manual monitoring entirely.

◆ *Pro Tip: Use the Problem → What you did → Result → What makes it different structure. Specificity wins.*

7 Tell me about a challenge you faced.

Tests: Problem-solving ability and resilience

When building the Crowd Control system, real-time performance was a serious challenge. YOLO detection is compute-heavy, and running it frame-by-frame was causing lag that made the system unusable for actual monitoring. I analyzed the bottleneck — the issue was that I was running full detection on every frame. I optimized by sampling frames at strategic intervals and applying centroid tracking between detections, which drastically reduced compute load without losing accuracy. The result: the system ran smoothly in real time, the dashboard updated live, and the entry restriction logic triggered correctly when thresholds were breached. That experience taught me that performance optimization isn't an afterthought — it's part of the design.

◆ *Pro Tip: Use a real, specific technical challenge. Vague answers like 'it was hard but I worked it out' don't impress engineers.*

8 Where do you see yourself in 5 years?

Tests: Ambition, clarity, and long-term fit

In five years, I see myself as a strong individual contributor — someone who's shipped complex systems, mentored junior engineers, and has a clear architectural instinct. I want to go deep in backend engineering or applied AI/data systems, not wide and shallow. I'm not chasing titles. I'm chasing impact — the kind that comes from deeply understanding a domain and solving its hardest problems. I want to be the engineer that people come to when something is genuinely difficult to build.

◆ *Pro Tip: Be specific about direction, not just titles. 'I want to be a senior engineer who builds X' is stronger than 'I want to grow.'*

9 Are you a team player?

Tests: Collaboration, communication, and attitude

Absolutely — and I'll give you a concrete example. In my AI-based Crowd Control project, I was the UI/UX developer and analyst working alongside teammates handling the YOLO model and backend logic. My role required me to understand their outputs well enough to build a dashboard that made sense of live data in real time. I didn't wait for handoffs — I stayed in sync, asked the right questions, and designed the UI around the actual data flow, not assumptions. The project worked because the team communicated tightly and each person took ownership of their piece while staying invested in the whole.

◆ *Pro Tip: Don't just say 'yes.' Give a team scenario that shows how you specifically contributed to shared success.*

10 Do you have any questions for us?

Tests: Your genuine interest and preparation

Yes, I have a few thoughtful ones: 1. What does the onboarding experience look like for a new engineer in this team — how quickly are new hires expected to contribute to production? 2. How does the team approach code reviews and design discussions? I want to know how technical decisions are made collaboratively. 3. What's the biggest engineering challenge the team is currently working through? I'd love to understand where my skills could have the most immediate impact. 4. How does the company support continuous learning — are engineers encouraged to experiment with new technologies or approaches?

◆ *Pro Tip: Always prepare 3–4 genuine questions. It signals curiosity and seriousness. Avoid asking about salary or leaves in the first interview.*